

Laxmi Charitable Trust's m
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java (EJ)

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
Date of Assignment: 29-08-2026	Date/Time of Submission: 29-08-2026

PRACTICAL 8

AIM :- Implement the following EJB applications with different types of Beans

- 1) Develop a simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

CODE

index.jsp :-

```
<!DOCTYPE html>
<html>
<head>
  <title>Hit Count Demo</title>
</head>
<body>
<form action="ShowHitCountServlet" method="post">
  <input type="submit"
    value="Take Hit Count demo with EJB Session Singleton Bean"/>
</form>
</body>
</html>
```

STHitCntBean :-

```
package EJPracts;
import javax.ejb.Singleton;
@Singleton
public class STHitCntBean {
  private int count = 0;
  public synchronized int incrHitCount() {
    return ++count;
  }
}
```

CCClient :-

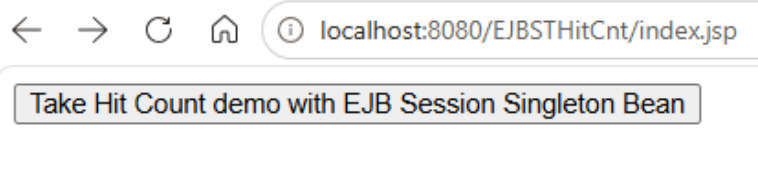
```
package servlet;
import EJPracts.STHitCntBean;
import java.io.IOException;
```

```

import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/ShowHitCountServlet")
public class CCClient extends HttpServlet {
    @EJB
    private STHitCntBean cntBean;
    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Hit Count</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>You visited this page: "
            + cntBean.incrHitCount()
            + " time/s</h1>");
        out.println("</body>");
        out.println("</html>");
    }
    @Override
    protected void doGet(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {
        doPost(request, response);
    }
}

```

OUTPUT :-



You visited this page: 3 time/s

- 2) Develop a simple visitor Statistics application using Message Driven Bean [MDB].

CODE :-

index.jsp :-

```
<%@page import="javax.jms.JMSEException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page import="javax.jms.Session"%>
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.TextMessage"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%!

    private static ConnectionFactory connFact;
    private static Queue queue;
    Connection conn = null;
    Session mysess = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;

%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Message Driven Bean</title>
</head>
<body style="color:gray;">
<h1>Welcome To Message Driven Bean</h1>
<%
try {
    InitialContext ctx = new InitialContext();
    queue = (Queue) ctx.lookup("jms/Queue");
    connFact =
        (ConnectionFactory) ctx.lookup("jms/QueueFactory");
    conn = connFact.createConnection();
    mysess = conn.createSession(
        false,
```

```

        Session.AUTO_ACKNOWLEDGE
    );
    msgProducer = mysess.createProducer(queue);
    msg = mysess.createTextMessage();
    msg.setText(request.getRemoteAddr());
    msgProducer.send(msg);
}
catch (JMSEException e) {
    System.out.println(
        "Exception occurred: " + e.toString()
    );
}
}
%>
</body>
</html>

```

VStatelessBean :-

```

package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private ResultSet rs = null;
    private Statement stmt = null;
    private String query = null;
    @PostConstruct
    public void connect() {
        try {
            Class.forName(
                "com.mysql.jdbc.Driver"
            ).newInstance();
            String connStr =
                "jdbc:mysql://localhost/reg";
            conn = DriverManager.getConnection(
                connStr,
                "root",
                "root"
            );\

```

```

        System.out.println(
            "Connection Success."
        );
    }
    catch (
        ClassNotFoundException |
        InstantiationException |
        IllegalAccessException |
        SQLException e
    ) {
        System.err.println(
            "Connection failed."
        );
    }
}
@PreDestroy
public void disconnect() {
    try {
        conn.close();
        System.out.println(
            "Connection closed."
        );
    }
    catch (SQLException e) {
        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    }
}
public void addVisitor(String host) {
    try {
        stmt = conn.createStatement();
        query =
            "INSERT INTO visitorstat "
            + "(host,visitcount) "
            + "VALUES('" + host + "',1)";
        stmt.executeUpdate(query);
    }
    catch (SQLException e) {
        try {
            stmt = conn.createStatement();
            query =
                "UPDATE visitorstat "

```

```

        + "SET visitcount = visitcount + 1 "
        + "WHERE host = '" + host + "'";
        stmt.executeUpdate(query);
    }
    catch (SQLException ex) {
        System.err.println(
            "Cannot execute update: "
            + ex.getMessage()
        );
    }
}
}
}
}

```

MsgDrvnBean :-

```

package EJPracts;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class MsgDrvnBean implements MessageListener {
    @EJB
    VStatelessBean vs;
    @Resource
    private MessageDrivenContext mdc;
    public MsgDrvnBean() {
    }
    @Override

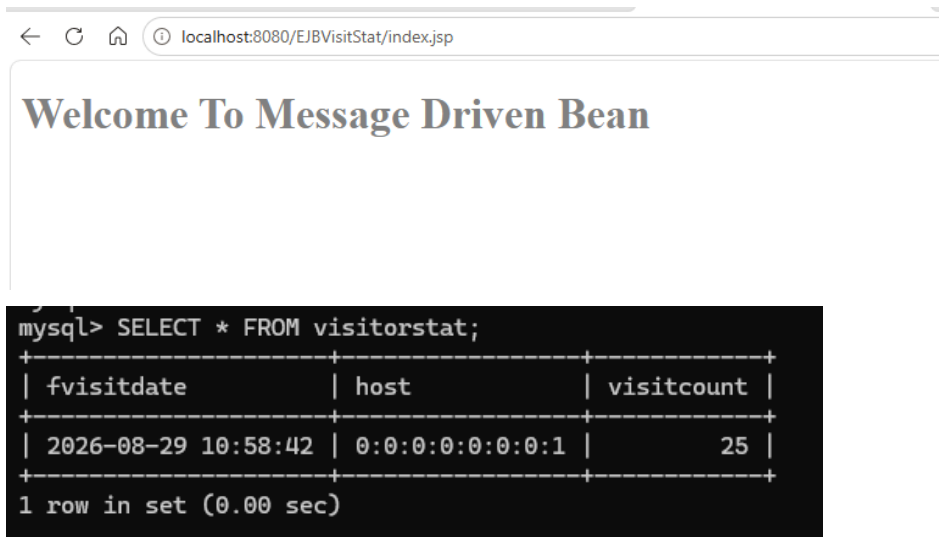
```

```

public void onMessage(Message message) {
    try {
        if (message instanceof TextMessage) {
            TextMessage msg =
                (TextMessage) message;
            vs.addVisitor(msg.getText());
        }
    }
    catch (JMSEException e) {
        mdc.setRollbackOnly();
    }
}
}

```

OUTPUT :-



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/EJBVisitStat/index.jsp'. The main content area of the browser shows the text 'Welcome To Message Driven Bean'. Below the browser window, there is a terminal window showing a MySQL query and its result.

```

mysql> SELECT * FROM visitorstat;
+-----+-----+-----+
| fvisitdate | host | visitcount |
+-----+-----+-----+
| 2026-08-29 10:58:42 | 0:0:0:0:0:0:1 | 25 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

- 3) Develop simple Marks Entry Application to demonstrate accessing Database using EJB.

CODE :-

index.jsp :-

```

<%@page import="java.util.ArrayList"%>
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPracts.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%!
    private static MarksEntryBean marksEntry;
    public void jspInit() {
        try {
            InitialContext ic = new InitialContext();
            marksEntry =
                (MarksEntryBean) ic.lookup(

```

```

        "java:global/EJBMarksEntry/MarksEntryBean"
    );
}
catch (Exception ex) {
    System.out.println(
        "Database Error" + ex.getMessage()
    );
}
}
%>
<%
if (request.getParameter("insertMarks") != null) {
    String name;
    String marks1;
    String marks2;
    String marks3;
    name =
        request.getParameter("stuName");
    marks1 =
        request.getParameter("sub1");
    marks2 =
        request.getParameter("sub2");
    marks3 =
        request.getParameter("sub3");
    marksEntry.addMarks(
        name,
        marks1,
        marks2,
        marks3
    );
}
%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Marks Entry</title>
</head>
<body>
<h2 style="text-align: center;">
    Enter Details
</h2>
<form name="result" method="post">

```



```
<table>
<tr>
  <td>Enter Students Name</td>
  <td>
    <input type="text"
      name="stuName"
      value="" />
  </td>
</tr>
<tr>
  <td>Enter Subject 1 Marks</td>
  <td>
    <input type="text"
      name="sub1"
      value="" />
  </td>
</tr>
<tr>
  <td>Enter Subject 2 Marks</td>
  <td>
    <input type="text"
      name="sub2"
      value="" />
  </td>
</tr>
<tr>
  <td>Enter Subject 3 Marks</td>
  <td>
    <input type="text"
      name="sub3"
      value="" />
  </td>
</tr>
<tr>
  <td colspan="2">
    <input type="submit"
      value="insertMarks"
      name="insertMarks" />
  </td>
</tr>
</table>
</form>
</body>
</html>
```

MarksEntryBean :-

```
package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.List;
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {
    List<String> details;
    String stuName;
    String sub1;
    String sub2;
    String sub3;
    private Connection conn = null;
    private ResultSet rs;
    private Statement stmt = null;
    private String query = null;
    public void createConnection() {
        try {
            Class.forName(
                "org.gjt.mm.mysql.Driver"
            ).newInstance();
            String connStr =
                "jdbc:mysql://localhost/reg";
            conn =
                DriverManager.getConnection(
                    connStr,
                    "root",
                    "root"
                );
        }
        catch (
            ClassNotFoundException |
            IllegalAccessException |
            InstantiationException |
            SQLException e
        ) {
            System.err.println(
                "Database Error." +
                e.getMessage()
            );
        }
    }
}
```

```

    }
}
public void addMarks(
    String stuname,
    String marks1,
    String marks2,
    String marks3) {
    createConnection();
    try {
        stmt = conn.createStatement();
        query =
            "INSERT INTO marks"
            + "(stuName,sub1,sub2,sub3)"
            + "VALUES("
            + stuname + ","
            + marks1 + ","
            + marks2 + ","
            + marks3 + ")";
        stmt.executeUpdate(query);
    }
    catch (SQLException e) {
        System.err.println(
            "Failed to insert." +
            e.getMessage()
        );
    }
}
}
}

```

OUTPUT :-

Enter Details

Enter Students Name

Enter Subject 1 Marks

Enter Subject 2 Marks

Enter Subject 3 Marks

```

mysql> SELECT * FROM marks;
+----+-----+-----+-----+-----+
| id | stuName   | sub1 | sub2 | sub3 |
+----+-----+-----+-----+-----+
|  1 | Saqlain Khan | 67   | 68   | 69   |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)

```